

ABHISHEK KS

Bachelor of Computer Applications

Community Institute of Commerce and Management Studies, Bangalore

+91 8310879483 | abhi2005aku@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI/ML enthusiast with hands-on experience in computer vision and full-stack AI development. Proficient in Python, OpenCV, Flask, React, and LLM-based systems, with a strong focus on building practical, real-world solutions

INTERNSHIP

Ai4sees Private Ltd

Jan 2026-May 2026 ([Link](#))

AI Research and Development Intern

- Developed an Unsupervised Anomaly Detection Framework using a **Transformer-** based **Teacher-Student architecture**, Leveraging knowledge distillation to identify structural and logical deviations in visual data.
- Trained and fine-tuned **ResNet50** backbones on custom-curated datasets, achieving high classification accuracy by Optimizing feature extraction layers for **specific patterns**.
- Evaluated and benchmarked **state-of-the-art** models including GLACE and EndoSLAM via Github integrated testing Environments identifying performance bottlenecks in endoscopic and medical imaging sequences.

PROJECTS

AI-Powered Image Restoration & Enhancement

([Link](#))

Technologies : ESRGAN (Enhanced Super-Resolution GAN), Streamlit, TensorFlow Hub, asynchronous resource caching (st.cache_resource)

- Built an **end-to-end image restoration application** using the **ESRGAN (Enhanced Super-Resolution GAN)** architecture to upscale and recover fine textures in low-resolution and degraded images.
- Optimized system performance by utilizing **asynchronous resource caching (st.cache_resource)**, reducing model load times and improving the overall user experience during real-time processing.
- Implemented a **high-performance computer vision pipeline using TensorFlow Hub**, converting raw image data into normalized tensors for efficient deep learning inference.

Friday – J.A.R.V.I.S Voice AI Assistant

([Link](#))

Technologies : Groq Whisper, edge-tts, ChromaDB, SentenceTransformers, WebRTC VAD.

- Built a voice assistant with an end-to-end pipeline covering **speech detection (WebRTC VAD)**, Transcription (**Groq Whisper large-v3-turbo**), LLM inference (**Groq LLaMA 3.1-8b streaming**), and **neural-benchmarked**.
- Engineered a **RAG** knowledge base using all-MiniLM-L6-v2 **sentence embeddings Semantic retrieval** over custom markdown knowledge files at query time

GalaxyEye Multi-Agent System

([Link](#))

Technologies : n8n, Docker, Cloudflare Tunnel, Gemini API, Linux.

- Designed a multi-agent routing workflow in **n8n** that dispatches user messages to specialized agents **Chat, Coding, Task, and LLM Reasoning** enabling domain-specific handling within a single interface
- Deployed self-hosted n8n in a **Dockerized** container on Linux with persistent volume storage and basic-auth security, ensuring reproducible setup and stateful workflow execution across restarts.

Open Source Contribution

- **widemem-ai** — Authored expanded configuration reference (per-class tables, runnable snippets for OpenAI/Anthropic/Ollama/local); PR merged by maintainer. Added decay-invariant test coverage for the scoring module (covering bounds, monotonicity, and step boundaries).
- **CodeGraphContext** — Resolved assigned issue #623 by aligning database config docs with CLI defaults (DEFAULT_DATABASE=falkordb) and updating CLI help text; validated via mkdocs --strict.
- **Causal-Memory-Layer** — Contributed a plain-language audit findings glossary (R1–R4 codes with examples) and a research-notes navigation map to improve onboarding for new readers.

Technical Skills and Interests

Languages: Python, Java, C (Basics).
Libraries and Frameworks: TensorFlow, Scikit-learn, Keras, Data visualization, FastAPI, MCP, LangChain, LangGraph.
Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Docker, HuggingFace.
Specialization: Machine Learning, Deep Learning, Generative AI, Natural Language Processing (NLP), Prompt Engineering, LLM, RAG.
Other Skills: Cybersecurity, Linux, Virtual Machine, n8n.

Certificates

- Bharatiya Antariksh Hackathon 2025 - ISRO** [\(Link\)](#)
- Worked on the problem statement: “Generating High-Resolution Lunar Digital Elevation Models (DEMs)”, involving lunar imagery processing and elevation reconstruction techniques.
- NMIT HACKS 2025 - NMIT** [\(Link\)](#)
- Worked on a Unified Chain ID framework designed to create a single verifiable identity across applications. Explored data linking, secure authentication, and workflow interoperability using modern backend/API approaches.

Education

Community Institute of Commerce and Management Studies, Bangalore <i>Bachelor of Computer Applications</i>	2023-2026 <i>CGPA: 7.7</i>
--	--------------------------------------